

ENV 6106

Air Pollution Modeling

Spring 2023

Topics

- BPIPPRM
- Expectation
 - Understand how BPIPPRM works and be able to run BPIPPRM
- Readings:
 - BPIP user guide
 - Addendum to the ISC3 User's Guide containing BPIPPRM documentation

Components of AERMOD

- AERMET: meteorological data preprocessor
 - AERSURFACE: for surface characteristics
 - AERMINUTE: for 1-min ASOS data
- AERMAP: terrain data preprocessor
- AERMOD: uses input data (meteorological, terrain data etc) to perform refined analysis
 - BPIPPRIM: a multi-building dimensions program incorporating the GEP technical procedures for building downwash applications.

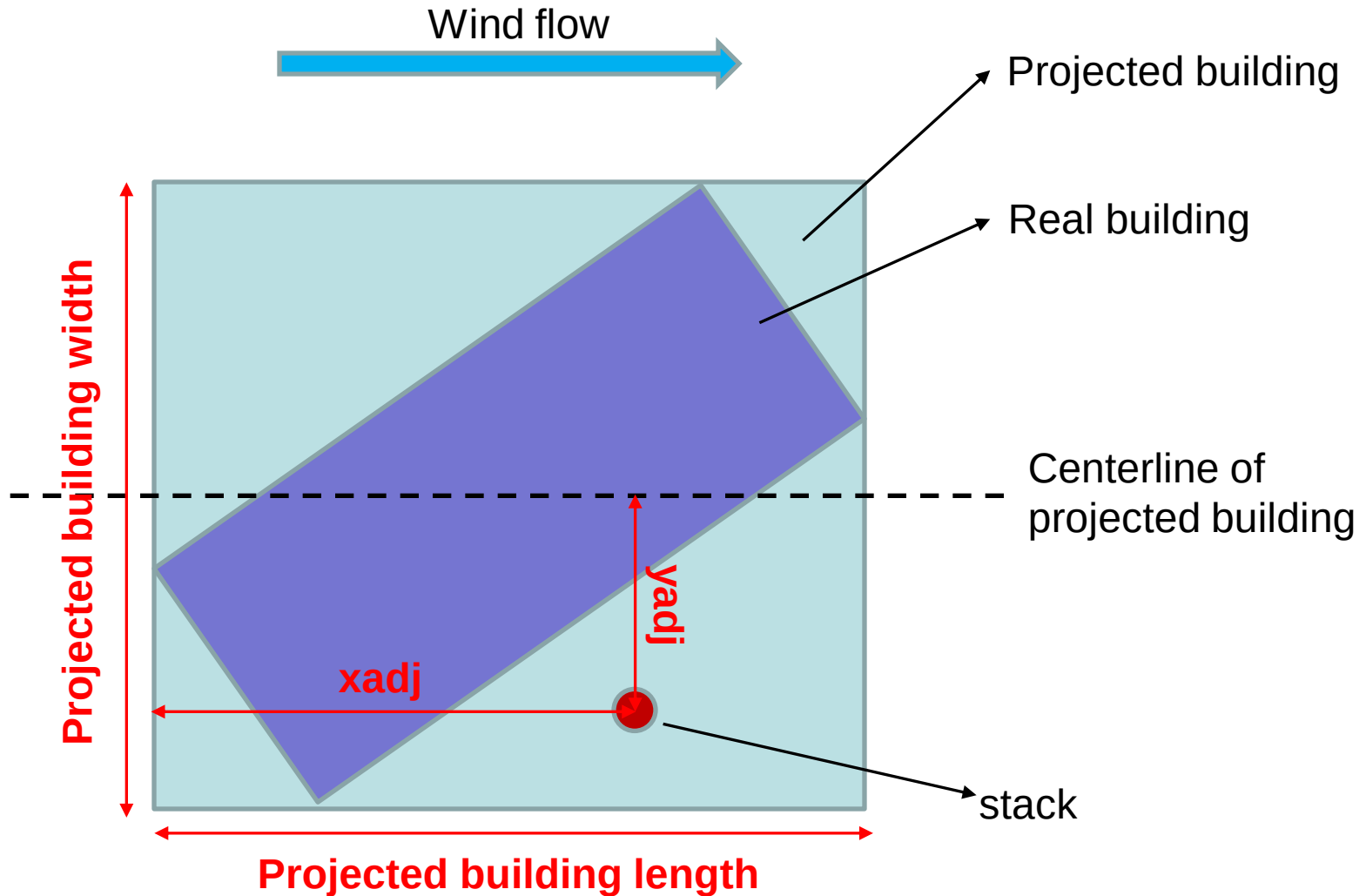
BPIPPRM

- Building Profile Input Program for PRIME
 - PRIME (plume rise model enhancements) is the building downwash model for AERMOD
- Downwash: As wind blows past blunt objects, flow disturbances occur, and low pressure zones are created. Depending on wind speed, vortexes and eddies are formed. These eddies can trap stack releases and bring them quickly to the ground

BPIPPRM

- BPIPPRM rotates around a stack, calculates projected building height, length and width, and relative positions of the stack to projected building, at different flow sectors
 - 36 directions in 10 degree increment
- Complicated building geometries can be specified in BPIPPRM
 - Strange shape, multi-story etc

BPIPPRM estimated params



BP1PPRM control file

<i>Title, quoted text</i>	'BPIP users guide test case #1'
<i>Model type</i>	'P'
<i>Unit and conversion</i>	'METERS' 1.00
<i>UTM flag and north angle</i>	'UTMN', 210.
<i># building</i>	1
<i>Building ID, tiers # and elev</i>	'L-Shape' 1 13.00
<i># corner and tier height</i>	6 26
<i>X and Y of corners</i>	-10. -20.
<i>....</i>	-10. 80.
<i>....</i>	40. 80.
<i>....</i>	40. 30.
<i>....</i>	90. 30.
<i>....</i>	90. -20.
<i># stacks</i>	4
<i>Stack name, elev, hgt, XY</i>	'Stk100' 11.00 25.00 -10.00 -20.00
<i>....</i>	'Stk101' 12.00 25.00 164.00 159.00
<i>....</i>	'Stk102' 13.00 25.00 136.00 121.00
<i>....</i>	'Stk103' 14.00 25.00 118.00 103.00

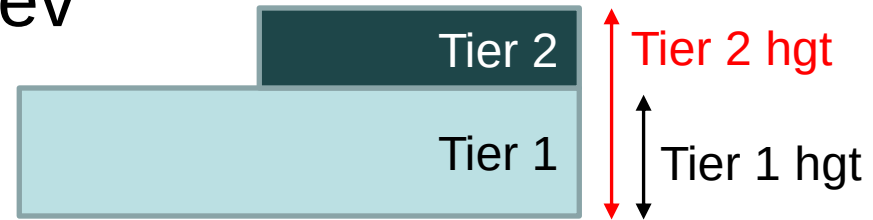
Test case A1P of BP1PPRM

Control file

- Model type: Quoted, can be
 - ‘P’: PRIME for AERMOD: output all params
 - ‘NP’: No PRIME - output projected building height and width
 - ‘ST’ & ‘LT’: for ISCST & ISC2LT models (no longer used)
- Unit & conversion
 - Unit of input numbers and conversion factor to meter
- UTM flag and north angle
 - ‘UTMN’: use local coordinates; ‘UTMY’: use UTM coordinates
 - Use UTM coordinates is recommended
 - Angle between facility north to true north
 - Input values may be based on rotated coordinates

Control file

- Building ID, tiers # and elev
 - Tier: how many levels
 - Ground elevation
 - Mean sea level or local elevation
 - Same elevation reference should be consistently used
- # corner and tier height
 - # corner points for current tier and height of tier
 - Repeat next tier after current tier done
- Stack name, elev, hgt, XY
 - Stack base elevation, stack height



BPIPPRM

- BPIPPRM runs differently, should be executed from command line
 - `bpipprm control_file output_file summary_file`
 - *output_file*: output parameters, paste to AERMOD run
 - *summary_file*: summary of calculation for each direction
 - Or use a batch file instead of command line